

CHAPTER IV SYSTEM DESIGN

System design is the process of the development of system specifications based on the results of the system analysis that has been done. The class diagram and interfaces design are the two important points in the application described in the system design.

During this phase, the source code and database design of Diet Project will be discussed in detail. The overview of the codes will be displayed in Class Diagram format, while the database design will be presented in ERD (Entity Relationship Diagram) and the structure table view.

4.1.Database Design

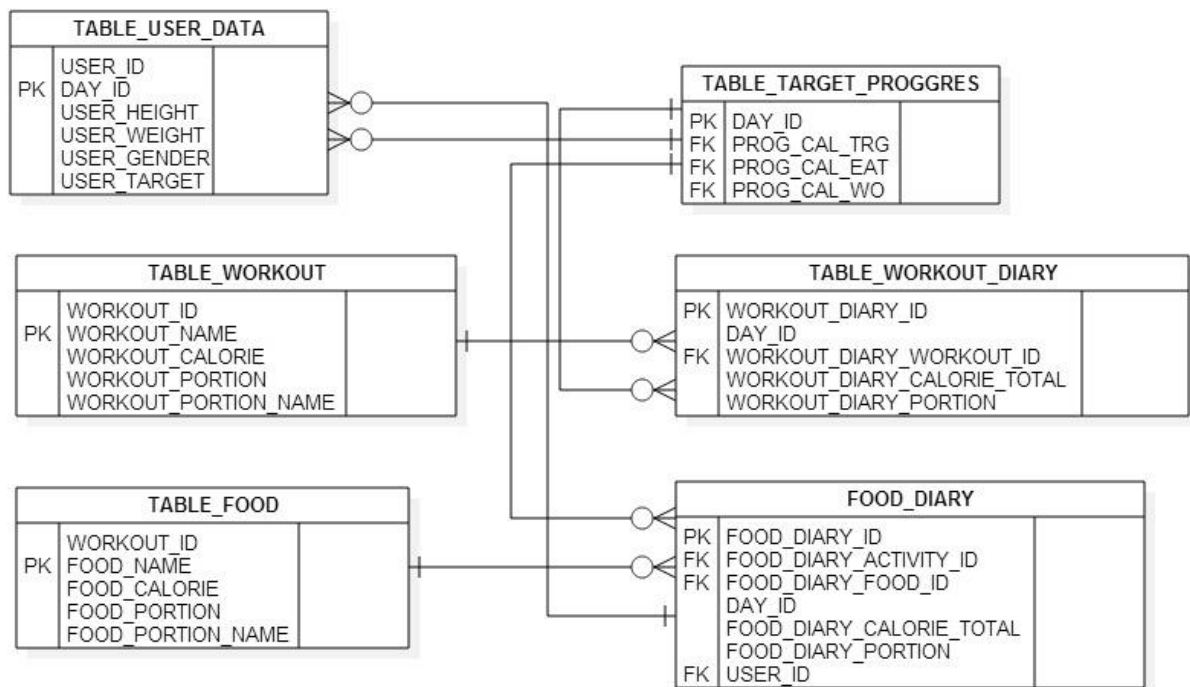


Figure 4.1 Entity Relationship Diagram of Diet Project

The Database design of DietProject is quite simple. There are only total of 6 table each are User Data, Target progress, Workout, Workout diary, Food, and Food Diary. The relationship is between food and its respective diary Tables. The relationship is one to many means that each entry of in food table can be accessed by many entry in Food diary table but each entry in food diary only able to have one food.

4.2. Class Diagram

A class diagram is a description of the object types in the system. It is used to demonstrate the properties and the operations of the class and object in the system

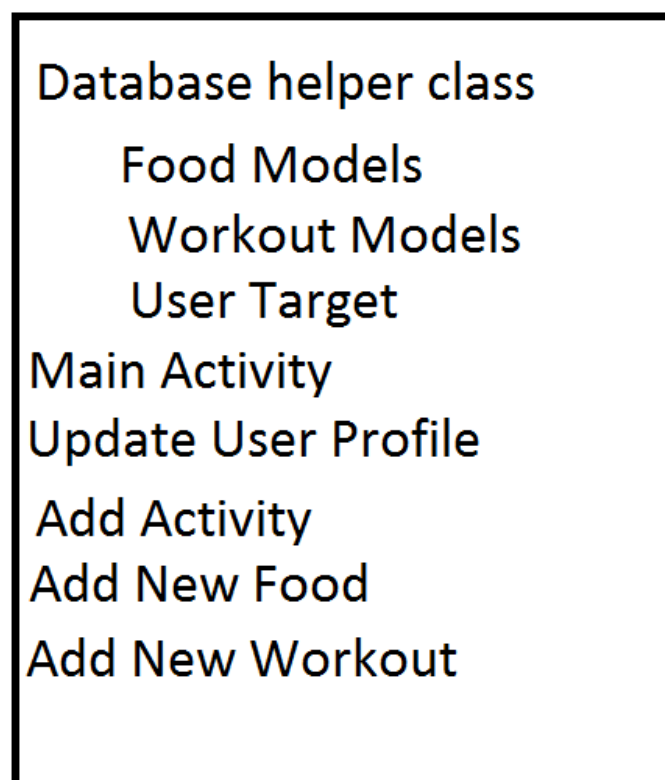


Figure 4.2 Class diagram of Diet Project software.

There are total of 19 classes in Diet Project. 1 Database helper classes used to call when the database being accessed by the program. 5 models classes uses to

create data models to be processed by other classes and to be saved in the databases.

And 13 classes is for activity classes.

4.1.1. Class Diagram of Diet Project

The Diet Project Is consists of several Java classes that was put in several packages and folders. As we can see at Figure 4.2 is the general layout of the DietProject Structure. As depicted, the project contains some folders and files. The AndroidManifest.xml contain the manifest of the program including Target android platform, etc. The Model Package is to contain Classes that will be used for the model of the database. It is connected to the helper class. The helper class is the gateway and Helper in aspect of trying to connect between databases and content.

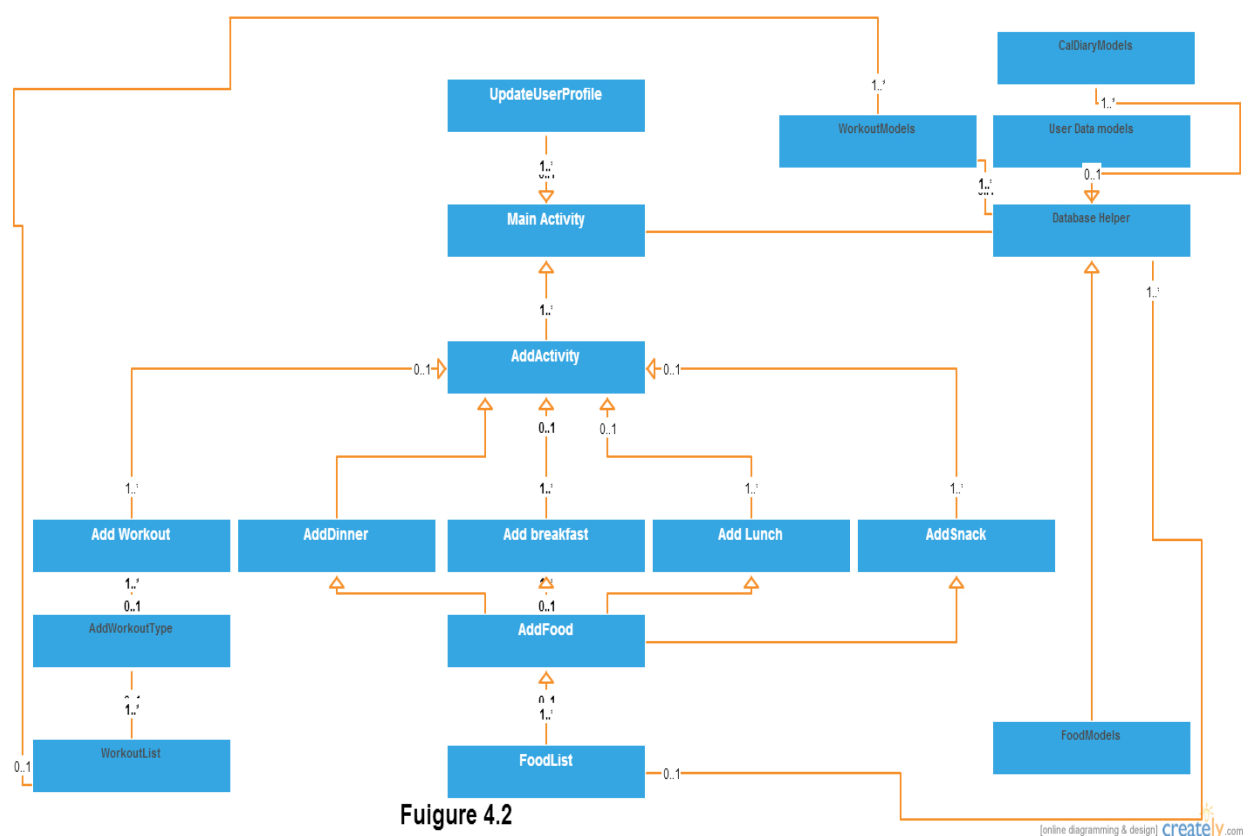


Figure 4.2

Figure 4.3 Class Diagram of Diet Project

The Class diagram in Figure 4.3 describes the Class in the *DietProject*. As the diagram stated Main Activity is the center and the first class to be loaded when starting the application.

Alongside of Main Activity is the Database helper class that become the gateway of database-related operations of the applications.

The classes inside the packages in Figure 4.3 will be described further in the next section. Each folder may have one or more classes and each will have their own function that makes the system works properly.

Figure 4.4 will show us classes extend from Main activity

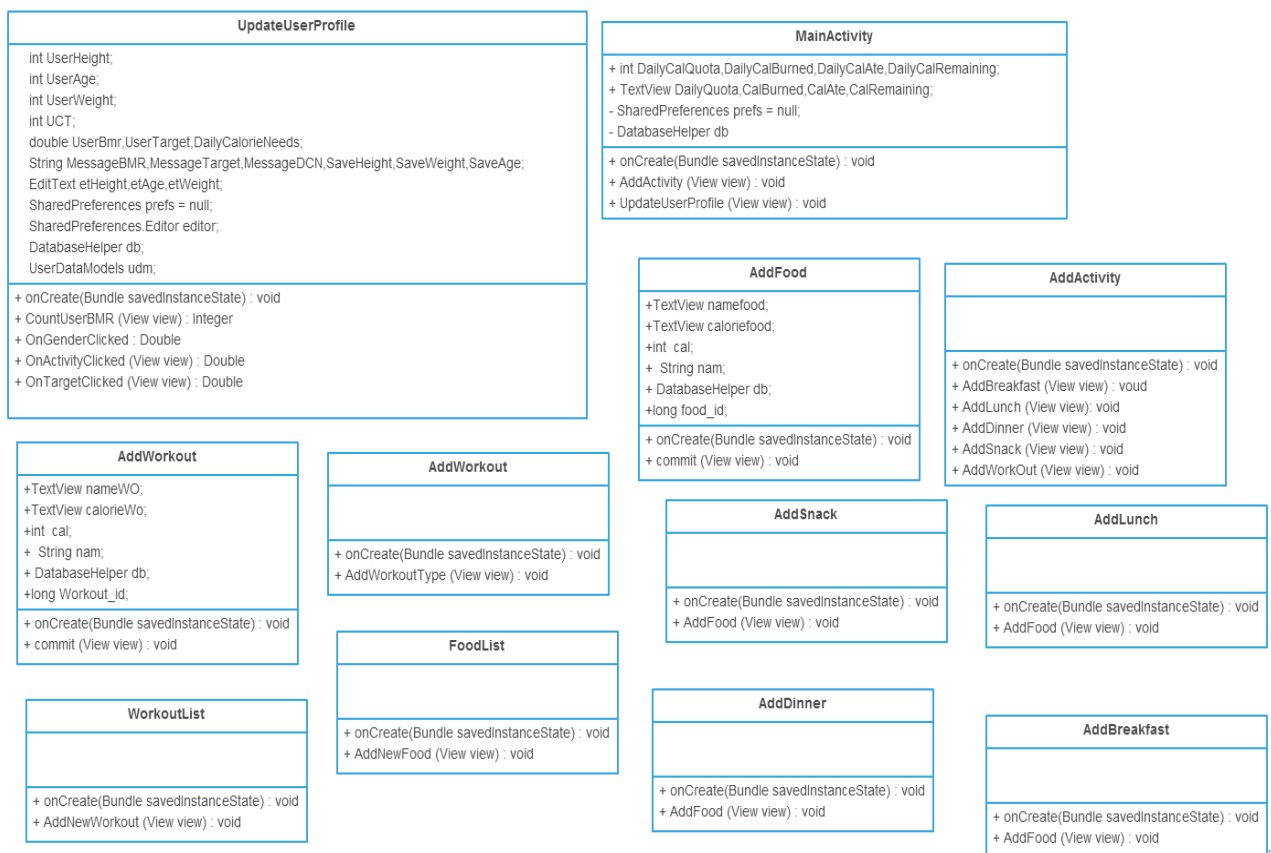


Figure 4.4 Classes extended from Main Activity

As can be seen in Figure 4.3 there are 12 classes that extended from main activity class. There are Update User Profile, Add Food, Add Activity, Add Workout, Add Workout type, Add Snack, Add Lunch, Workout List, Food List, Add Dinner, and Add breakfast. All of these classes will be explained in the following sections.

4.1.2. Update User Profile

Table 4.1 show us Update user profile class. This Class is used to change User data

UpdateUserProfile
<pre>int UserHeight; int UserAge; int UserWeight; int UCT; double UserBmr, UserTarget, DailyCalorieNeeds; String MessageBMR, MessageTarget, MessageDCN, SaveHeight, SaveWeight, SaveAge; EditText etHeight, etAge, etWeight; SharedPreferences prefs = null; SharedPreferences.Editor editor; DatabaseHelper db; UserDataModels udm;</pre>
<pre>+ onCreate(Bundle savedInstanceState) : void + CountUserBMR (View view) : Integer + OnGenderClicked : Double + OnActivityClicked (View view) : Double + OnTargetClicked (View view) : Double</pre>

Table 4.1 Update User Profile Class

of the User. In here also affected the layout of User Data Activity. Here User can change Height, Weight Age Gender and Their Target. This class also accounted to return user BMR and user daily calorie target. The resulted User BMR and daily calorie target then sent to main activity class for further processing. In this class user are asked to fill out a form asking their personal data like age, height, weight, gender, activity level, and user target. Using these input we can count BMR, Daily

calorie needs, and Daily Calorie Target. All that will be saved in Shared Preferences and can be used later at main activity.

4.1.3. Main Activity

Table 4. 2 shows us the class of Main Activity class is the first class that got called by the application. It will start on Create () process to check whether is the program is first time to be run or not. It also checking and assign existing shared preferences data to layout field in order to show it to the user.

MainActivity
+ int DailyCalQuota,DailyCalBurned,DailyCalAte,DailyCalRemaining; + TextView DailyQuota,CalBurned,CalAte,CalRemaining; - SharedPreferences prefs = null; - DatabaseHelper db
+ onCreate(Bundle savedInstanceState) : void + AddActivity (View view) : void + updateUserProfile (View view) : void

Table 4.2 Main Activity Class

4.1.4. Database Helper

There is only one Java class that handle The Database. Called database helper file in this package. Figure 4.6 shows us the Schema of Database Helper Class. Database Helper will be the class that will be called the most when running the program. The rest of Class that connected to Database Helper mainly are only a models class. Which means it only contain variables and getter-setter method. These classes act like they are a table in a database. Model class enables us to get object from database or to inject object into database.

DatabaseHelper
+ FoodModels : FoodModels + ID : Integer - Name: String - Calorie : Integer
+ createFood (FoodModels food) : food_id + getFood (long food_id) :FoodModels + getAllFood () : ArrayList<FoodModels> + getAllFoodSpecial () : ArrayList<Map<String, String>> + updateFood (FoodModels foodModels) : Integer + deleteFood (long food_id) : Unspecified + createWorkout (WorkoutModels Workout) : Workout_id + getWorkout (long Workout_id) :WorkoutModels + getAllWorkout () : ArrayList<WorkoutModels> + getAllWorkoutSpecial () : ArrayList<Map<String, String>> + updateWorkout (WorkoutModels WorkoutModels) : Integer + deleteWorkout (long Workout_id) : Unspecified + createUserData (UserDataModels UserData) : UserData_id + getUserData (long UserData_id) :UserDataModels + getAllUserData () : ArrayList<UserDataModels> + getAllUserDataSpecial () : ArrayList<Map<String, String>> + updateUserData (UserDataModels UserDataModels) : Integer + deleteUserData (long UserData_id) : Unspecified + createDiary (DiaryModels Diary) : Diary_id + getDiary (long Diary_id) :DiaryModels + getAllDiary () : ArrayList<DiaryModels> + getAllDiarySpecial () : ArrayList<Map<String, String>> + updateDiary (DiaryModels DiaryModels) : Integer + deleteDiary (long Diary_id) : Unspecified + createFoodList (FoodListModels FoodList) : FoodList_id + getFoodList (long FoodList_id) :FoodListModels + getAllFoodList () : ArrayList<FoodListModels> + getAllFoodListSpecial () : ArrayList<Map<String, String>> + updateFoodList (FoodListModels FoodListModels) : Integer + deleteFoodList (long FoodList_id) : Unspecified + putData(String name, Integer calorie) : HashMap<String, String>

Table 4.3 Database Helper Class

4.3.User Interface Design

Interface design is an essential part of the program. This interface is required so the system can communicate with the user who wants to use the program. Nowadays, the most users friendly User Interface Design is Graphical User Interface (GUI).

GUI is easy to use and looks attractive to the eye. Based on that concept, this project will also use the GUI as the User Interface Design.

The main concept of DietProject's GUI is simple and informative. The aim is to provide a simple interface but rich of information, so it will be easy to use and provide a good amount of information.

Since Diet Project is mainly aimed to People who wants to start to live healthier, the concept above is the most suitable design of this project. A simple interface also acceptable by those who already having healthy live and want a quick reference from the program.

The design of Diet Project interface will be discussed further in the next section.

4.3.1. Home Screen

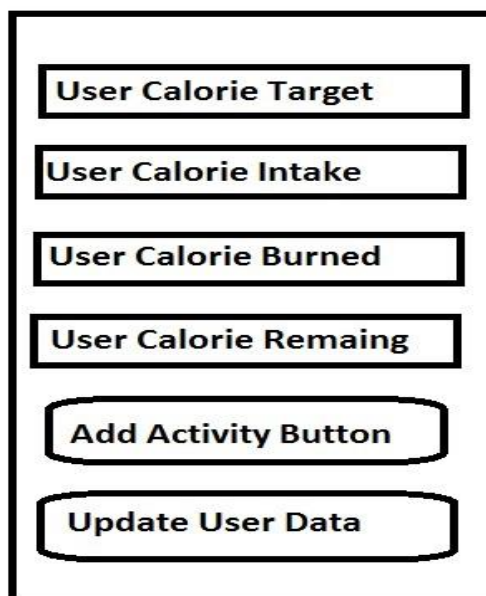


Figure 4.4 Design of Home Screen

The Home Screen will provide a one stop information for all of the Applications.

Home Screen will consists of 4 Text Field each will show User's calorie target, Calorie Intake, Calorie Burned, and remaining User's calorie.

Also there is two buttons there are Add Activity button and Update User Data.

Add Activity Button used for adding activity likes foods and workout.

Update User Data used to update previously submitted user data

4.3.2. Update User Data

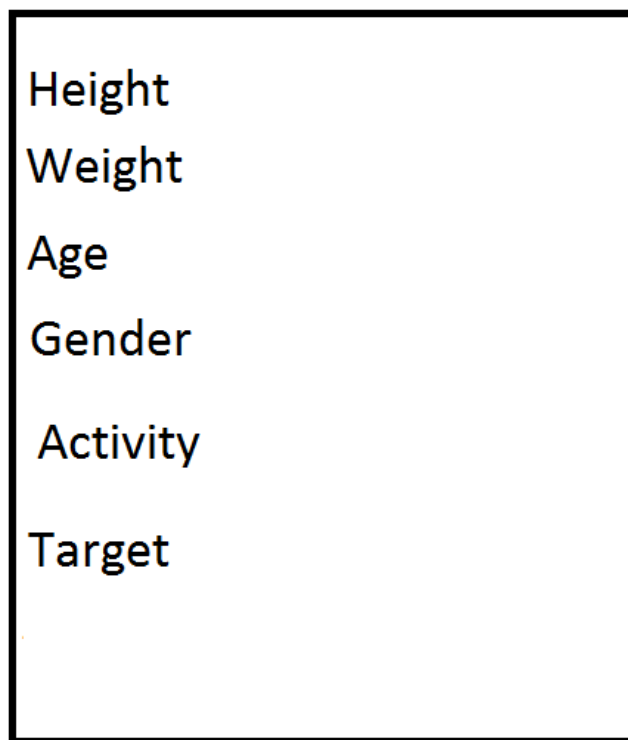
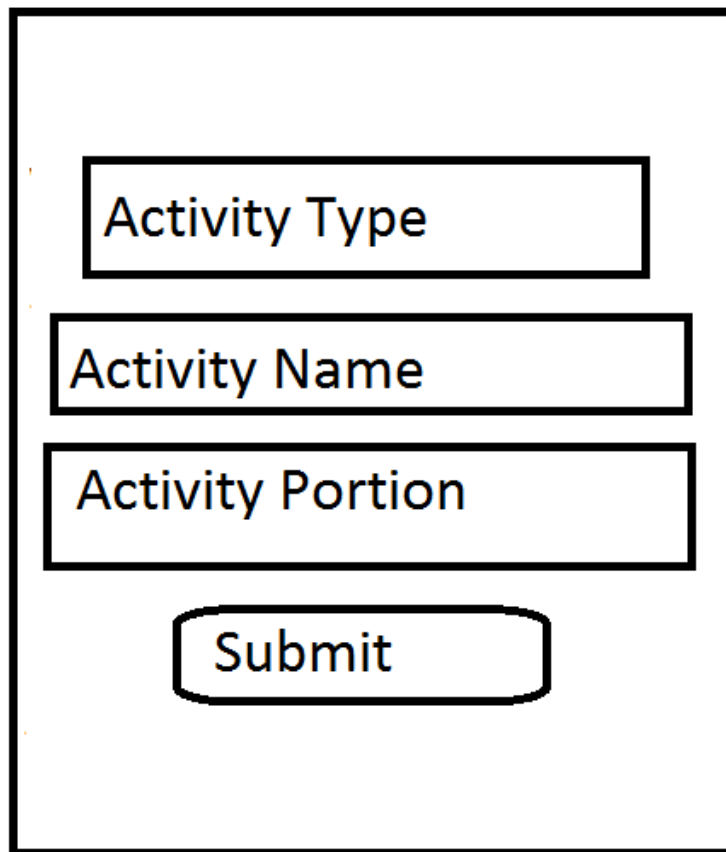
A rectangular box representing a user profile update form. Inside the box, the following labels are listed vertically from top to bottom: Height, Weight, Age, Gender, Activity, and Target. Each label is positioned to the left of an empty space, presumably for a text input field.

Figure 4.5 Update user Profile Interface design

Update User Data screen will appears at two occasions

1. First Time launch of the applications
2. When User choose Update User data buttons

If there is already saved user data. The column will be filled with latest saved information. User able to fill out their data and update it as long as they think it necessary. In Update User profile Also User able to change their target and



The diagram illustrates the 'Add activity' interface design. It consists of a large rectangular container with a black border. Inside this container, there are four input fields stacked vertically, each with a black border and placeholder text. The first field is labeled 'Activity Type', the second 'Activity Name', and the third 'Activity Portion'. Below these three fields is a rounded rectangular button labeled 'Submit'.

Figure 4.6 Add activity interface design

activity level. Because of this Applications is intended for only one person so there is no need to save personal info like names or ID.

4.3.3. Add Activity

Add Activity gives the user chance to add new activity to the applications.

There are 6 choice to choose with.

1. Breakfast
2. Lunch

3. Dinner
4. Snack
5. Work Out

Each will lead the user to the respective page that will show all activity they already had and the possibility to add new activity.

4.3.4. Food / workout list

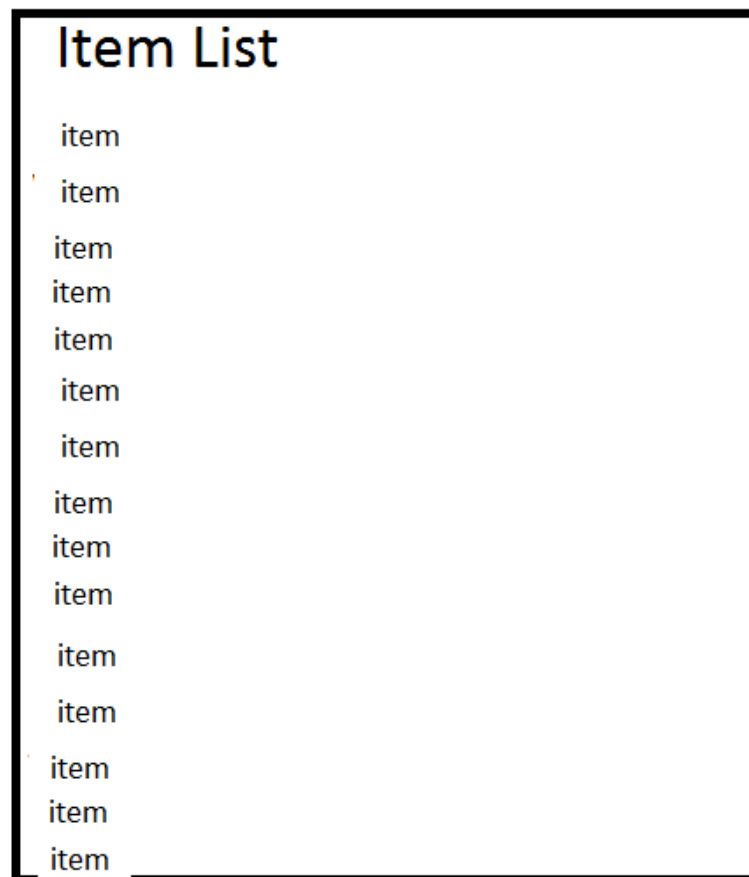


Figure 4.7 Food / Workout list Interface design

In case the user can't find the intended items user have the options to add new items. This display will appear when a User choose to add new item to the database eaten Food or Work Out. Using this display user can easily choose the Item they want to add to their activity. It provide user with list of item name with

calorie amount. Users can easily select the food or workout they want and then it will be added to the activity page.

4.3.5. Add new item

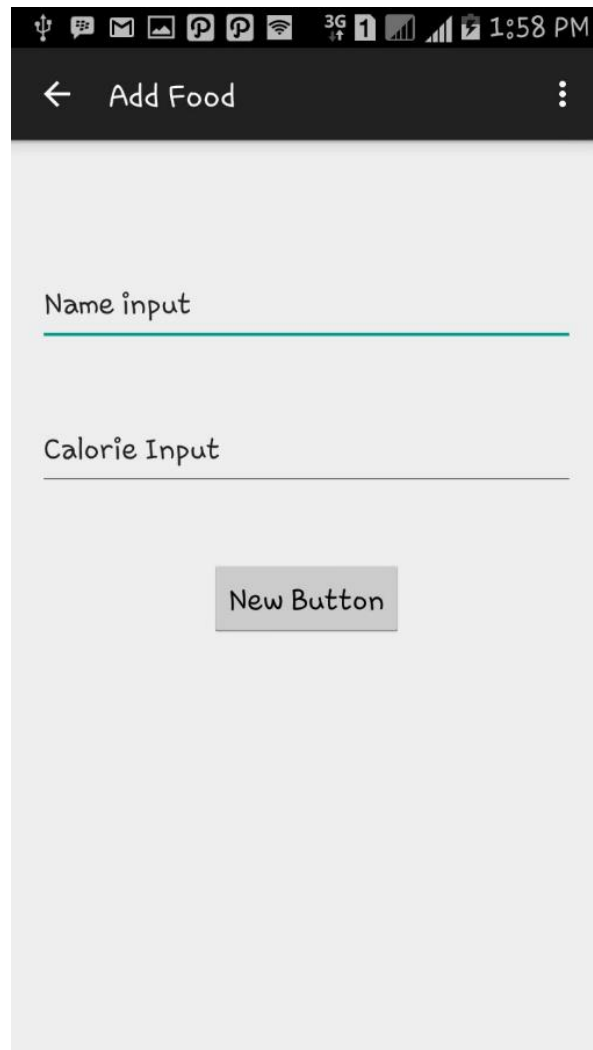
The image shows a mobile application interface for adding a new food item. At the top, there is a dark header bar with a back arrow, the text 'Add Food', and a menu icon. Below the header, the main content area is light gray. It contains two text input fields: 'Name input' and 'Calorie Input', each with a horizontal line underneath. At the bottom center, there is a gray button labeled 'New Button'. The top of the screen shows a status bar with various icons and the time '1:58 PM'.

Figure 4.8 Add new item interface design

In case the Database didn't contain the items User's trying to look at the applications give the options for users to add their data by their own.